AGENTIC AI ENGINEERING

Assignment 2 - Safe and Production-Aware Student Support Agent

Extend the same agent with privacy, permission control, human approval and auditability.

PGDM (Big Data Analytics / Artificial Intelligence & Data Science) | Batch 2025-27

Assignment Description

Continue the same Student Support Agent from Assignment 1. Do not start a new project. The objective is to make the existing agent safer and more suitable for real institutional use.

Expected final output: the same chat-style application from Assignment 1, now able to protect student data, block unauthorized access, request approval for sensitive actions and record important actions.

What to Add

- Student identity: the system should know which student is currently signed in.
- Data privacy: a student may access their own private record but must not be able to read or modify another student's private data.
- Authorization must be enforced in normal application / tool code, not only through an LLM instruction.
- At least one sensitive action should require confirmation or human approval before it is executed.
- Important actions should create an audit record showing who requested the action, what was attempted, whether it was allowed or approved, and the final result.
- The tracing and evaluation from Assignment 1 should remain functional after these controls are added.

Required Privacy and Safety Scenarios

- Logged-in student asks for their own fee or scholarship status - allowed.
- Logged-in student asks for another student's private record - blocked.
- Student requests a sensitive change to their own record - approval required before execution.
- Approval is rejected - the action must not execute.
- Sensitive information such as an OTP or password is provided in chat - it must not become persistent memory.

Final Evaluation

Run a final test set and compare the result with Assignment 1. The evaluation should report measurable results.

Measure	What to show
Task accuracy / success	Does the agent still complete normal support tasks correctly?
Privacy tests	Were attempts to access another student's data blocked?
Unsafe-action tests	Did sensitive actions wait for approval instead of executing automatically?
Latency	Approximate response time after adding privacy and approval controls.
Token usage	Input and output token usage for the tested conversations.
Trace / audit evidence	Show the important tool calls, permission decision, approval status and result.

Submission

- Updated working Student Support Agent.
- A short explanation of the privacy and approval rules implemented.
- Sample trace / audit records from the required scenarios.
- A final evaluation table comparing key measurements with Assignment 1.

The assignment focuses on correct agent behaviour. Authentication, approval and audit mechanisms should be implemented clearly and should be explainable during the demonstration.